FRANKLIN YESUTOR FERGUSON

UAV Systems Engineer and Drone Survey · Data & GIS Platform Specialist · Aerospace Engineer

CONTACT

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franklin's Portfolio

Mastiff ST GM-046-1558
Madina, Accra, Ghana

EDUCATION

BSc Aerospace Engineering
First Class Honours
KNUST, Kumasi, Ghana
Aug 2018 – Sep 2022

Google Cybersecurity Cert.
Coursera · Jan 2025

CORE COMPETENCIES

SURVEY AND MAPPING

- RTK/PPK Survey Setup
- Drone-Based Site Surveys
- Orthomosaic Generation
- Topographic Mapping & DTM/DSM
- Stockpile and cut/fill volumes
- GCP layout and accuracy

UAV & AVIATION

- UAV Design & Fabrication
- Aerostat Operations
- Aerial Survey & Photogrammetry
- Agricultural UAV Spraying
- UAV Maintenance & Repair
- Multispectral & LiDAR Imaging

DATA, MODELS & SYSTEMS

- 3D Modeling: Metashape & Pix4D
- GIS Mapping: QGIS
- LiDAR, Multispectral & Thermal Data
- Data Analytics & Automation: Power BI, Power Apps, SQL & Python
- SolidWorks · Fusion 360 · Siemens NX
- Power BI Dashboards
- SQL & Data Modelling
- Python Analytics

PROFESSIONAL PROFILE

Drone survey and geospatial professional with a first-class degree in aerospace engineering and four years of turning the ground into data that decision-makers trust. Since 2022 I have mapped farmland across twelve regions of Ghana, an entire municipal district and, today, the active pits of Newmont's Ahafo South Mine, where I play a critical role in the planning, coordination, and execution of UAS and aerostat operations. I take every survey from first control point to final deliverable, producing surfaces, volumes and maps built to engineering accuracy and trusted for operational decisions.

PROFESSIONAL EXPERIENCE

UAV & Aerostat Technician

Jan 2026 - Present
Current Role

Isen Ventures (Contracted to Newmont Ghana)
Ahafo Mine, Ghana

- **Survey & mapping**: Run aerial and GNSS surveys end to end, from RTK/PPK control to orthomosaics, DTMs, contours and volumes in Metashape, Pix4D and QGIS, environment and security.
- **Aerostat surveillance:** command live missions on the RT SkyStar 180 tethered aerostat, from deployment and payload sensor operation to real-time monitoring and safe recovery.
- **Threat intelligence & Intelligence :** Detect and classify illegal mining in real time with EO/IR, multispectral sensors and identify potential illegal access routes, briefing Newmont's command, Security and the MRU.
- **UAV operations:** Fly and maintain a hybrid VTOL fixed-wing UAV and multirotors for mapping, inspection and BVLOS corridor surveillance between Ahafo South and Ahafo North.
- **Digital systems:** Built a Power BI command dashboard, a Power App for flight and maintenance records, and a Node.js tool that turned paper inspection checklists into automated reports.
- **Training & Capability:** Designed and delivered a mapping and survey course for the aerostat team (Aug 2026), and train operational teams in flight and data capture.

Team Lead, Research and Development / UAV Engineer

SKT Live Limited Westlands, Accra, Ghana
Jan 2023 - Dec 2025

- Survey & mapping: Led nationwide UAV survey, spraying and environmental monitoring operations, including national-scale mapping campaigns for financial and government clients.
- Multi-sensor capture: Captured and processed RGB, multispectral, infrared and LiDAR data into orthomosaics, terrain models and cadastral boundary maps.

PROFESSIONAL EXPERIENCE (CONTINUED)

- Engineering & R&D: Designed and built a modular seed-dispersal system for the TT Aviation 30L, pioneering UAV-assisted afforestation in Ghana. Led SolidWorks and Fusion 360 prototyping that cut hardware failures.
- Data & analytics: Built SQL, Python and Power BI dashboards and KPI frameworks for the Tree Crops Development Authority.
- Training: Drone Instructor with Flying Labs, mentoring students in UAV technology.

Chief Operating Officer (COO)

Feb 2022 - Dec 2023

Wingxtra Aerospace Limited

Co-founded during final year of undergraduate study; led the company full-time following graduation in September 2022.

Accra, Ghana

- Led end-to-end UAV product development — design, prototyping, testing, and production of drone systems for surveillance, agriculture, and precision operations.
- Directed financial planning, cash flow management, and daily operations of the UAV start-up managed full product lifecycle from concept through manufacturing to client deployment.
- Built and led cross-functional engineering teams to refine drone performance through rigorous testing cycles and technology capability expansion.

Aircraft Maintenance Intern

Jun 2019 - Jul 2021

School of Trade Training, Ghana Airforce Base, Accra

Accra, Ghana

Gained hands-on experience with aircraft systems, maintenance procedures, and airworthiness checks on active Ghana Air Force aircraft.

SELECTED PROJECTS & FIELD OPERATIONS

Ghana Eximbank · Ghana
SKT Live Limited · Ghana
Led nationwide UAV survey across 12 regions, 140 farms (~3,412 ha) capturing RGB, infrared, and LiDAR data to support crop health analysis, yield estimation, and agricultural investment risk assessment.

Ga East Municipal Assembly · Ghana
SKT Live / Arkadiah (Singapore)
Partnered with Arkadiah (Singapore) to survey and map all properties, businesses, and residents, enabling a digital revenue management system for property tax and compliance enforcement.

Dedekebe Land Survey · Ghana (~1,000 ha)
SKT Live Limited · Ghana
Surveyed and mapped land parcels using the Wingtra ONE fixed-wing VTOL drone, delivering high-resolution orthomosaics, cadastral boundary mapping, and geospatial analysis

UNDP Afforestation Initiative · Mauritania
SKT Live Limited · Ghana
Designed and built a modular UAV seed dispersal system using Arduino-controlled mechanisms for automated reforestation operations.

REFERENCES

Available upon request.